WORKED SOLUTION

STEP III 2014, Question 7

Pure • Vectors and Cyclic Quadrilaterals

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Read the question before you go on. It is in the STEP Support Programme's question database:

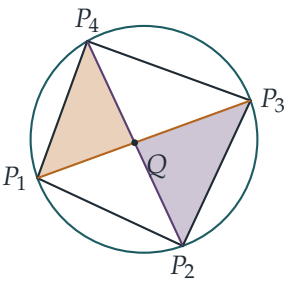
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The key idea

Part (i) is the intersecting chords theorem, and it comes from angles in the same segment. The vector parts give the same fact a second time. Four points in a plane are "affinely dependent", and the coefficients in that dependence locate the crossing point of the diagonals, so they divide each diagonal in a known ratio. Putting those ratios into (i) gives the final relation.

Solution

Two facts about the configuration are used throughout. First, no three of the P_i are collinear, because a line meets a circle in at most two points. Second, since the vertices are labelled in order round the circle, P_1 and P_3 lie on the two different arcs cut off by the chord P_2P_4. So they are on opposite sides of the line P_2P_4, and the diagonals cross at a point Q strictly inside both segments P_1P_3 and P_2P_4.



Part (i)

In the triangles P_1QP_4 and P_2QP_3:

- $\angle P_1QP_4 = \angle P_2QP_3$, since they are vertically opposite angles at Q;

- $\angle QP_1P_4 = \angle P_3P_1P_4$ and $\angle QP_2P_3 = \angle P_4P_2P_3$. These are equal because both are angles subtended by the chord P_3P_4 at points P_1, P_2 on the same arc (the arc from P_4 to P_3 through P_1 and P_2).

So the triangles are similar, with $P_1 \leftrightarrow P_2$, $Q \leftrightarrow Q$, $P_4 \leftrightarrow P_3$. Corresponding sides are in proportion:

$$\frac{P_1Q}{P_2Q} = \frac{QP_4}{QP_3}, \qquad \text{so} \qquad (P_1Q)(QP_3) = (P_2Q)(QP_4),$$

as required.

Part (ii)

The cyclic quadrilateral lies in a plane, so the three vectors $\mathbf{p}_2 - \mathbf{p}_1$, $\mathbf{p}_3 - \mathbf{p}_1$, $\mathbf{p}_4 - \mathbf{p}_1$ all lie in a two-dimensional space. Three vectors in two dimensions are linearly dependent, so there are numbers b_2, b_3, b_4, not all zero, with

$$b_2(\mathbf{p}_2 - \mathbf{p}_1) + b_3(\mathbf{p}_3 - \mathbf{p}_1) + b_4(\mathbf{p}_4 - \mathbf{p}_1) = \mathbf{0}.$$

(Concretely: $\mathbf{p}_2 - \mathbf{p}_1$ and $\mathbf{p}_3 - \mathbf{p}_1$ are not parallel, since P_1, P_2, P_3 are not collinear, so they span the plane and $\mathbf{p}_4 - \mathbf{p}_1$ is a combination of them. Then we may take $b_4 = -1$.) Now put $a_1 = -(b_2 + b_3 + b_4)$ and $a_i = b_i$ for $i = 2, 3, 4$. Then $\sum a_i = 0$, not all a_i are zero, and

$$\sum_{i=1}^{4} a_i\mathbf{p}_i = a_1\mathbf{p}_1 + b_2\mathbf{p}_2 + b_3\mathbf{p}_3 + b_4\mathbf{p}_4 = b_2(\mathbf{p}_2 - \mathbf{p}_1) + b_3(\mathbf{p}_3 - \mathbf{p}_1) + b_4(\mathbf{p}_4 - \mathbf{p}_1) = \mathbf{0},$$

so $(*)$ holds.

Part (iii)

$a_1 + a_3 \neq 0$. Suppose instead that $a_1 + a_3 = 0$. Then $a_2 + a_4 = 0$ too, by the first equation of $(*)$, so $a_3 = -a_1$ and $a_4 = -a_2$. The second equation becomes

$$a_1(\mathbf{p}_1 - \mathbf{p}_3) + a_2(\mathbf{p}_2 - \mathbf{p}_4) = \mathbf{0}.$$

The lines P_1P_3 and P_2P_4 are different (otherwise all four points would be collinear) and they meet at Q, so they are not parallel. Hence $\mathbf{p}_1 - \mathbf{p}_3$ and $\mathbf{p}_2 - \mathbf{p}_4$ are non-zero and not parallel, which forces $a_1 = a_2 = 0$. Then all four a_i are zero, a contradiction. So $a_1 + a_3 \neq 0$, and likewise $a_2 + a_4 = -(a_1 + a_3) \neq 0$.

The intersection point. From $(*)$, $a_1\mathbf{p}_1 + a_3\mathbf{p}_3 = -(a_2\mathbf{p}_2 + a_4\mathbf{p}_4)$ and $a_1 + a_3 = -(a_2 + a_4)$. Dividing one by the other,

$$\mathbf{r} = \frac{a_1\mathbf{p}_1 + a_3\mathbf{p}_3}{a_1 + a_3} = \frac{a_2\mathbf{p}_2 + a_4\mathbf{p}_4}{a_2 + a_4}.$$

The first form shows $\mathbf{r} = \mathbf{p}_1 + \dfrac{a_3}{a_1 + a_3}(\mathbf{p}_3 - \mathbf{p}_1)$, a point of the line P_1P_3. In the same way the second form shows it is a point of the line P_2P_4. Two non-parallel lines meet in only one point, so $\mathbf{r}$ is the position vector of Q.

Ask yourself How do we turn this formula for Q into the lengths P_1Q and QP_3?

Subtract. Using $\mathbf{q} = \mathbf{r}$,

$$\mathbf{q} - \mathbf{p}_1 = \frac{a_3(\mathbf{p}_3 - \mathbf{p}_1)}{a_1 + a_3}, \qquad \mathbf{p}_3 - \mathbf{q} = \frac{a_1(\mathbf{p}_3 - \mathbf{p}_1)}{a_1 + a_3}.$$

Taking lengths gives $P_1Q = \dfrac{|a_3|}{|a_1 + a_3|}P_1P_3$ and $QP_3 = \dfrac{|a_1|}{|a_1 + a_3|}P_1P_3$. In the same way, from the second form of $\mathbf{r}$, $P_2Q = \dfrac{|a_4|}{|a_2 + a_4|}P_2P_4$ and $QP_4 = \dfrac{|a_2|}{|a_2 + a_4|}P_2P_4$.

The signs. Since Q lies strictly between P_1 and P_3, the fraction $\frac{a_3}{a_1+a_3}$ lies strictly between 0 and 1. Then $\frac{a_1}{a_1+a_3} = 1 - \frac{a_3}{a_1+a_3}$ is also strictly between 0 and 1. Two positive numbers have a positive product, so

$$\frac{a_1a_3}{(a_1 + a_3)^2} > 0, \qquad \text{that is, } a_1a_3 > 0.$$

The same argument on P_2P_4 gives $a_2a_4 > 0$. So $|a_1a_3| = a_1a_3$ and $|a_2a_4| = a_2a_4$.

The deduction. Substitute into part (i):

$$\frac{a_1a_3}{(a_1 + a_3)^2}(P_1P_3)^2 = (P_1Q)(QP_3) = (P_2Q)(QP_4) = \frac{a_2a_4}{(a_2 + a_4)^2}(P_2P_4)^2.$$

Since $(a_2 + a_4)^2 = (a_1 + a_3)^2 \neq 0$, multiplying through by it gives

$$a_1a_3(P_1P_3)^2 = a_2a_4(P_2P_4)^2,$$

which is the required result.

Checks

- Take the square with vertices $(1,0), (0,1), (-1,0), (0,-1)$. Then $\mathbf{p}_1 + \mathbf{p}_3 = \mathbf{0} = \mathbf{p}_2 + \mathbf{p}_4$, so $a = (1,-1,1,-1)$ satisfies $(*)$. The formula gives Q at the origin, and $a_1a_3(P_1P_3)^2 = 4 = a_2a_4(P_2P_4)^2$.
- For an unsymmetrical case, take $P_1, \ldots, P_4$ on the unit circle at angles $200°, 290°, 20°, 120°$, as in the diagram. All three claims of (iii), including $a_1a_3 > 0$ and $a_2a_4 > 0$, were confirmed by a computer check on this and on thousands of random cyclic quadrilaterals, to within 10^{-10}.
- The answer survives rescaling all the a_i by the same factor λ, as it must: both sides of the final relation scale by λ^2.

Common slips

- Pairing the wrong angles in (i): the equal angles at P_1 and P_2 stand on the chord P_3P_4, and this fixes which sides correspond.

- Dropping the modulus signs in (iii) without comment. The relation with absolute values is easy; the work is in showing $a_1a_3 > 0$ and $a_2a_4 > 0$, which uses the fact that Q is inside both diagonals.
- Forgetting why the diagonals are not parallel when proving $a_1 + a_3 \neq 0$.

Beyond STEP

The quantity $(P_1Q)(QP_3)$ is the *power* of the point Q with respect to the circle; it is the same for every chord through Q. Part (ii) is the statement that four points in a plane are affinely dependent. The numbers a_i are barycentric-style coordinates, and the formula for Q is a first look at how affine geometry handles points, lines and ratios without any coordinates. Strang's *Introduction to Linear Algebra* covers linear dependence and the dimension count behind (ii).

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